ROSHAN KUMAR MAHTO

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate with practical experience building Python applications and REST APIs. Developed a Gemini-powered text summarization backend using FastAPI, with API versioning, Pydantic validation, input sanitization, standardized responses, caching, rate limiting, health checks, and automated tests. Also built tested desktop and expense-management applications and am developing a React-based portfolio. Interested in entry-level Python Developer, Backend Developer, and AI Application Engineer roles.

TECHNICAL SKILLS

Programming: Python, SQL, Java, C, C++
Backend Development: FastAPI, REST API design, Pydantic, Uvicorn, API versioning, dependency injection
Python Concepts: Object-oriented programming, modular design, exception handling, input validation, debugging, virtual environments
AI & Data Libraries: Google Gemini API, NumPy, Pandas, Matplotlib, scikit-learn fundamentals
Testing & Quality: pytest, unit testing, fixtures, mocking, API testing, edge-case testing, code coverage
Frontend: React, JavaScript, HTML, CSS, Vite
Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Docker, Windows, Linux
Databases: MySQL, MongoDB

PROJECTS

AI TEXT SUMMARIZER | *Python • FastAPI • Gemini API • Pydantic • pytest • Docker • React • Vite*

- Built a versioned FastAPI backend for an AI-powered text summarization service, integrating the Gemini API through a dedicated service layer and supporting short, bullet-point, and detailed summary styles.
- Designed request and response schemas with Pydantic; added dependency injection and environment-based configuration to keep API logic modular and maintainable.
- Implemented text validation and sanitization, HTML stripping, normalization, standardized API responses, centralized error handling, health checks, request IDs, and processing-time headers.
- Added TTL-based in-memory caching and SlowAPI rate limiting to reduce repeated work and control request frequency.
- Created 25 passing pytest tests covering endpoints, input validation, health checks, mocked AI responses, service failures, error handling, and rate limiting. Backend is implemented; frontend development/integration remains in progress.

EXPENSE TRACKER | *Python • OOP • pytest • Git*

- Developed a modular expense-management application for recording, categorizing, updating, deleting, and reviewing expenses.
- Used reusable components, input validation, and exception handling; verified core behavior and edge cases with 17 passing automated tests.

PROFESSIONAL CALCULATOR | *Python • CustomTkinter • pytest • Git*

- Built a desktop calculator with a dark-themed interface, arithmetic operations, decimal and negative values, keyboard input, reset controls, and invalid-input handling.
- Separated calculation logic from the interface and tested arithmetic operations, invalid expressions, and division-by-zero behavior with 8 passing tests.

PORTFOLIOX — INTERACTIVE PORTFOLIO (IN DEVELOPMENT) | *React • JavaScript • Vite • HTML • CSS • Git*

- Developing a personal portfolio to showcase projects, technical skills, education, and professional links through reusable React components and responsive styling.
- Organizing the site into clear sections to present software projects and career information in a maintainable interface.

EDUCATION

B.Tech in Computer Science Engineering — Chandigarh University | 2021–2025 | CGPA: 6.3/10
Class XII, CBSE: 73.6% | Class X, CBSE: 76.4%

CERTIFICATIONS

- Developing AI Applications with Python and Flask
- Automate Cybersecurity Tasks with Python
- Introduction to Big Data with Spark and Hadoop

RELEVANT COURSEWORK & LANGUAGES

Coursework: Data Structures and Algorithms, DBMS, Operating Systems,
Computer Networks, Artificial Intelligence, Machine Learning
Languages: English, Hindi